

Teoria e Aplicação de Grafos

Roteiro da aula:

- **Métricas em Grafos**
- **Coeficiente de Aglomeração**
- **Exemplos**
- **Medidas de Centralidade**
- **Exemplos**

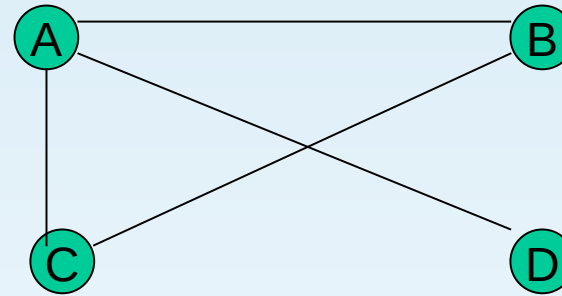
Métricas em Grafos

- **Medidas (métricas) típicas em grafos são:**
 - **Número de nós(vértices)**
 - **Número de arestas(elos)**
 - **Graus do vértices**
 - **Grau do grafo**
 - **Comprimento do menor caminho entre dois nós**
 - **Diâmetro: o maior dos menores caminhos entre dois nós de um grafo**
 - **+...**

Histogramas dos graus

Exemplo:

A tem grau 3, B tem grau 2,
C tem grau 2, D tem grau 1

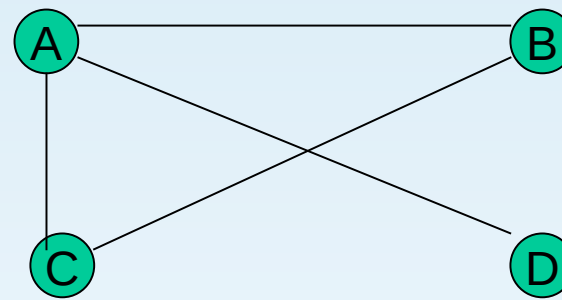


Outras métricas

Histogramas dos graus

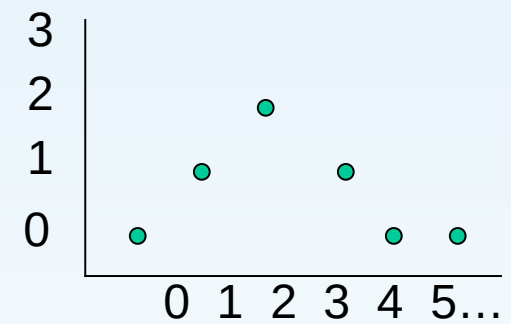
Exemplo:

A tem grau 3, B tem grau 2,
C tem grau 2, D tem grau 1



O *histograma dos graus* deste grafo é:

I.e. possui 0 nós de grau 0, 1 de grau 1,
2 de grau 2, 1 de grau 3, e 0 de grau >3.



Grau:	0	1	2	3	4	5
Frequência	0	1	2	1	0	0

Distribuição (de probabilidade) dos graus

Grau	0	1	2	3	4	5
Frequência	0	1	2	1	0	0
<i>Distribuição</i>	0	0.25	0.5	0.25	0	0

A distribuição do grau é uma função $P(k)$, que fornece a probabilidade de um nó, randomicamente escolhido de um grafo, ter grau k .

Distribuição (de probabilidade) dos graus

Questões pertinentes:

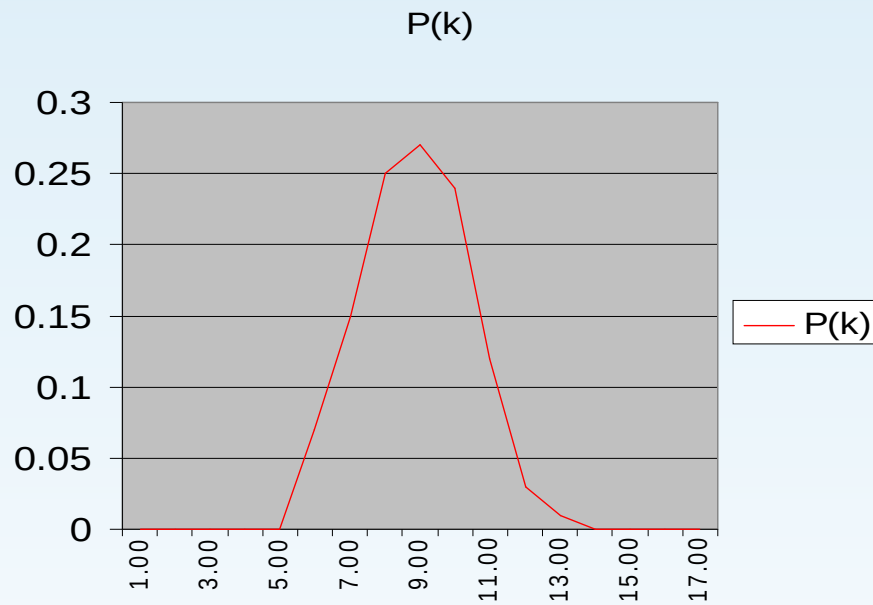
Em um grafo qualquer, dado o número de nós, como seria a distribuição de probabilidade dos graus?

Se as arestas fossem adicionadas randomicamente, uma por uma?

Como seria o grau médio dos nós?

Distribuição (de probabilidade) dos graus

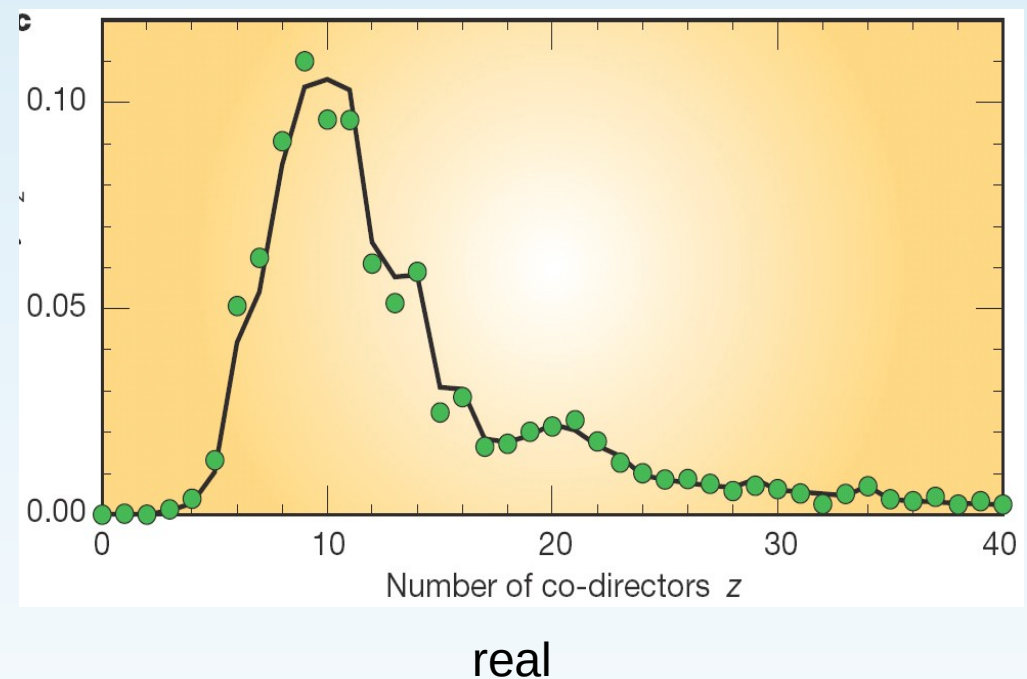
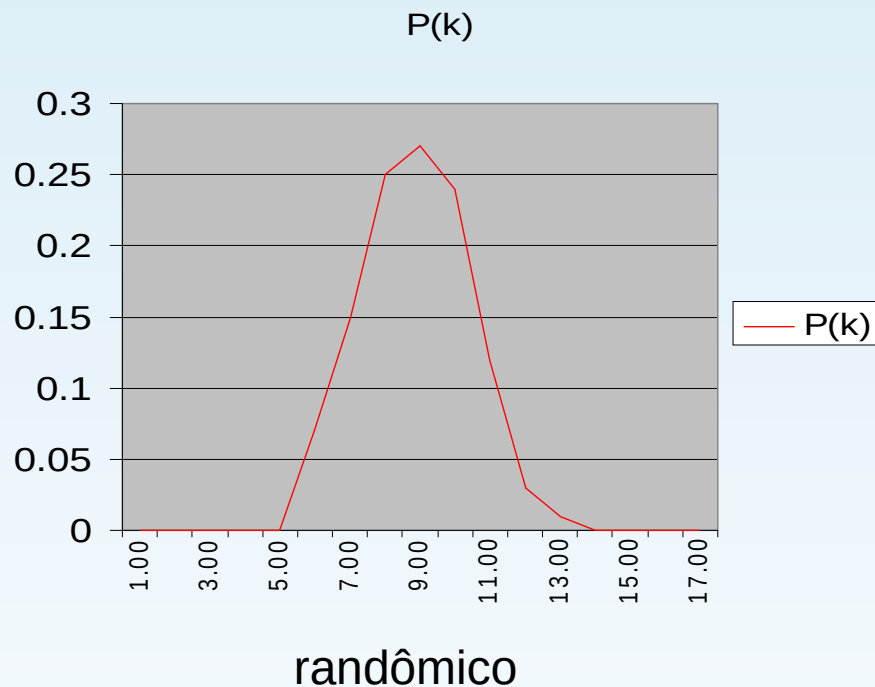
Exemplo:



Situação típica quando arestas são adicionadas randomicamente. Se houver n nós, e m arestas adicionadas, então o pico será em $2m/n$, o grau médio. Ou seja, para um nó randomicamente escolhido, o grau mais provável é o médio.

Distribuição (de probabilidade) dos graus

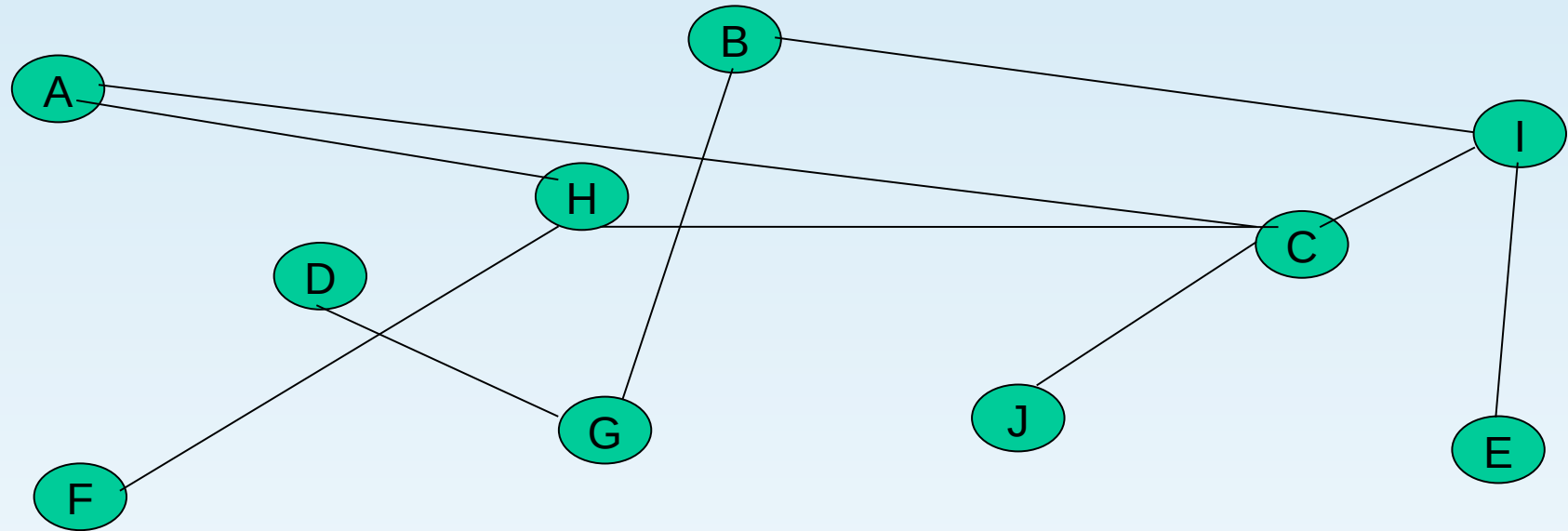
Diferente de grafos randômicos, grafos reais produzem distribuições com caudas longas



Na região da cauda, seguindo uma relação exponencial

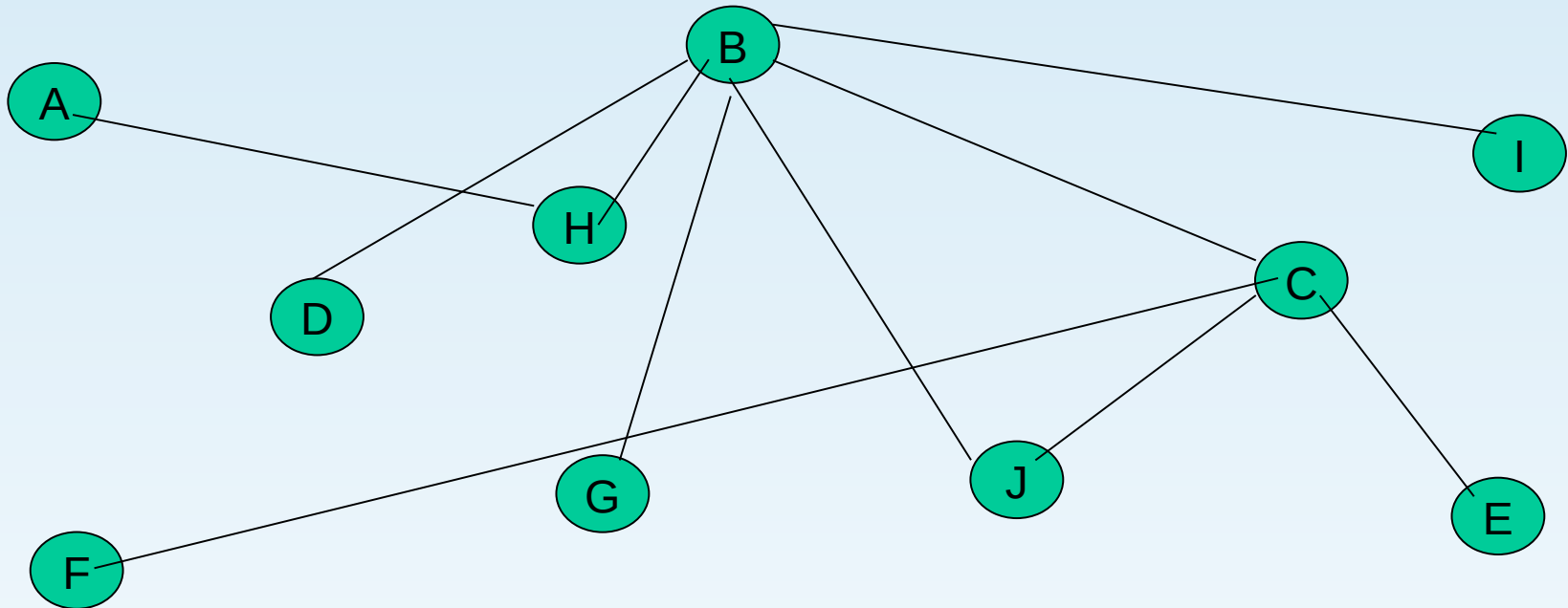
$$P(k) \approx k^{-\lambda} \Rightarrow \log(P(k)) \approx -\lambda \log(k)$$

Exemplo



Distribuição dos graus: 0.4 0.3 0.2 0.1
Maior caminho mais curto (diâmetro): DF 6

Exemplo



Distribuição dos graus: 0.6 0.2 0.0 0.1 0.0 0.1

Maior caminho mais curto (diâmetro): AE 4

Coeficiente de Aglomeração

Suponha que o nó i tenha n_i vizinhos.

Haverá portanto $n_i(n_i - 1)/2$

triângulos possíveis (arestas que ligam os vizinhos do nó i)

Suponha que t_i dessas arestas estejam no grafo.

O Coeficiente de Aglomeração do nó i é definido como:

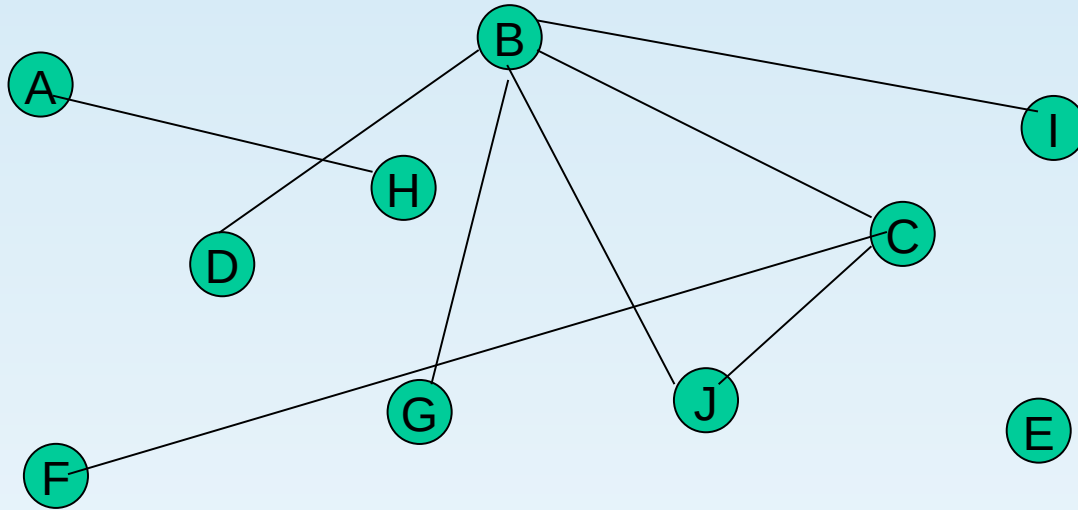
$$2t_i / n_i(n_i - 1)$$

A média desse valor para um grafo é chamada de Coeficiente de Aglomeração do grafo, i.e.:

$$C = 1/N \sum_{i=1}^N c_i$$

onde c_i é o coeficiente de aglomeração do nó i , e N é o número de nós do grafo.

Exemplo

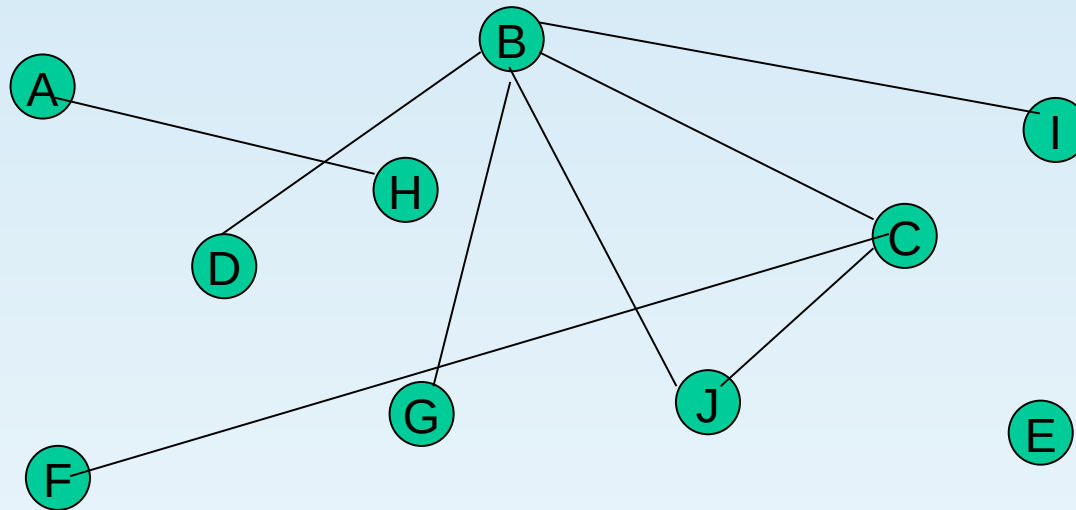


Considere nó B .

B possui 5 vizinhos : D, G, J, C, I

Cada par distinto de vizinhos (há $5 \times 4 / 2 = 10$ pares distintos) formariam triângulos *potenciais* com B . O triângulo $BJCB$ existe, porque a aresta CJ existe. Mas nenhum dos outros existem. O coeficiente de aglomeração de B é $1/10$.

Exemplo



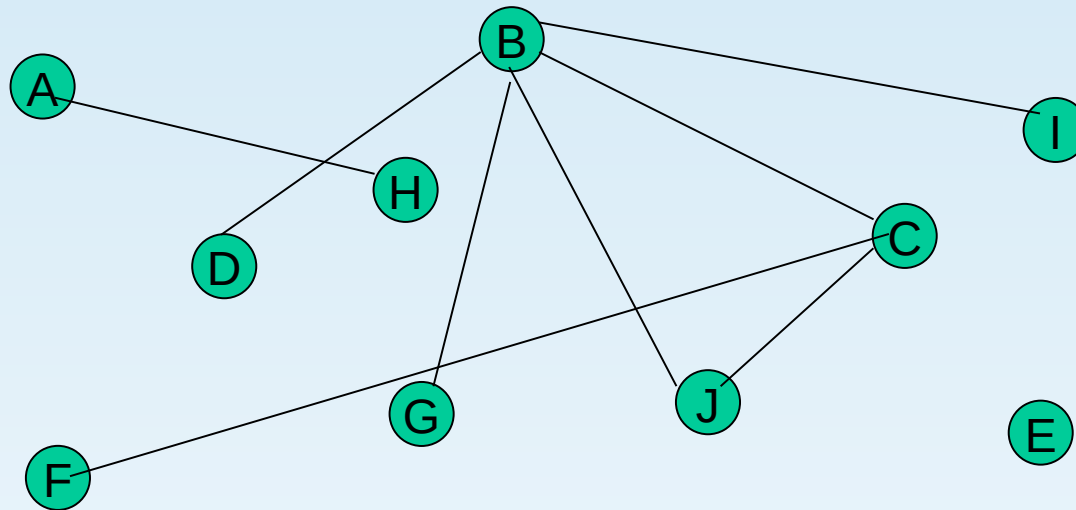
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Qual o coeficiente de aglomeração médio do grafo?

Exemplo



Considere nó B .

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Qual o coeficiente de aglomeração médio do grafo?

$$(0, 1/10, 1/3, 0, 0, 0, 0, 0, 0, 1) = (0,143)$$

Exemplo

TABLE I. The general characteristics of several real networks. For each network we have indicated the number of nodes, the average degree $\langle k \rangle$, the average path length ℓ , and the clustering coefficient C . For a comparison we have included the average path length ℓ_{rand} and clustering coefficient C_{rand} of a random graph of the same size and average degree. The numbers in the last column are keyed to the symbols in Figs. 8 and 9.

Network	Size	$\langle k \rangle$	ℓ	ℓ_{rand}	C	C_{rand}	Reference	Nr.
WWW, site level, undir.	153 127	35.21	3.1	3.35	0.1078	0.00023	Adamic, 1999	1
Internet, domain level	3015–6209	3.52–4.11	3.7–3.76	6.36–6.18	0.18–0.3	0.001	Yook <i>et al.</i> , 2001a, Pastor-Satorras <i>et al.</i> , 2001	2
Movie actors	225 226	61	3.65	2.99	0.79	0.00027	Watts and Strogatz, 1998	3
LANL co-authorship	52 909	9.7	5.9	4.79	0.43	1.8×10^{-4}	Newman, 2001a, 2001b, 2001c	4
MEDLINE co-authorship	1 520 251	18.1	4.6	4.91	0.066	1.1×10^{-5}	Newman, 2001a, 2001b, 2001c	5
SPIRES co-authorship	56 627	173	4.0	2.12	0.726	0.003	Newman, 2001a, 2001b, 2001c	6
NCSTRL co-authorship	11 994	3.59	9.7	7.34	0.496	3×10^{-4}	Newman, 2001a, 2001b, 2001c	7
Math. co-authorship	70 975	3.9	9.5	8.2	0.59	5.4×10^{-5}	Barabási <i>et al.</i> , 2001	8
Neurosci. co-authorship	209 293	11.5	6	5.01	0.76	5.5×10^{-5}	Barabási <i>et al.</i> , 2001	9
<i>E. coli</i> , substrate graph	282	7.35	2.9	3.04	0.32	0.026	Wagner and Fell, 2000	10
<i>E. coli</i> , reaction graph	315	28.3	2.62	1.98	0.59	0.09	Wagner and Fell, 2000	11
Ythan estuary food web	134	8.7	2.43	2.26	0.22	0.06	Montoya and Solé, 2000	12
Silwood Park food web	154	4.75	3.40	3.23	0.15	0.03	Montoya and Solé, 2000	13
Words, co-occurrence	460.902	70.13	2.67	3.03	0.437	0.0001	Ferrer i Cancho and Solé, 2001	14
Words, synonyms	22 311	13.48	4.5	3.84	0.7	0.0006	Yook <i>et al.</i> , 2001b	15
Power grid	4941	2.67	18.7	12.4	0.08	0.005	Watts and Strogatz, 1998	16
<i>C. Elegans</i>	282	14	2.65	2.25	0.28	0.05	Watts and Strogatz, 1998	17

Coeficiente para redes/grafos reais em geral é muito maior do que para redes randômicas de mesmo número de nós e arestas. (Albert & Barabasi, 02)

Exemplo

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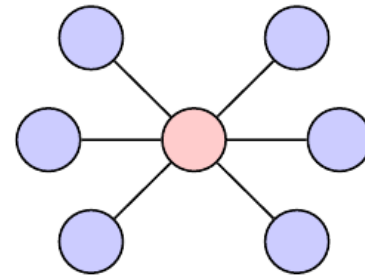
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Diâmetros de grafos grandes tendem a ser pequenos.

(Albert & Barabasi, 02)

Medidas de Centralidade

- ▶ Identify the **most important nodes** in a graph given its **topology**
 - ⇒ **Not** based on the **nature** of the particular node
- ▶ Different definitions of **importance** give rise to different **centrality measures**
 - ⇒ Degree, closeness, eigenvector, betweenness, Katz
 - ⇒ They induce a **centrality ranking** on the nodes
- ▶ Centrality measures are **widely used**
 - ⇒ Targeted marketing
 - ⇒ Network vulnerability to attacks
 - ⇒ Epidemiology control
 - ⇒ Power in exchange networks



(imagens/slides A. Ajorlou, MIT)

Centralidade de grau

- ▶ **Local** measure of the importance of a node within a graph
- ▶ Sum of the **weights of incident edges**

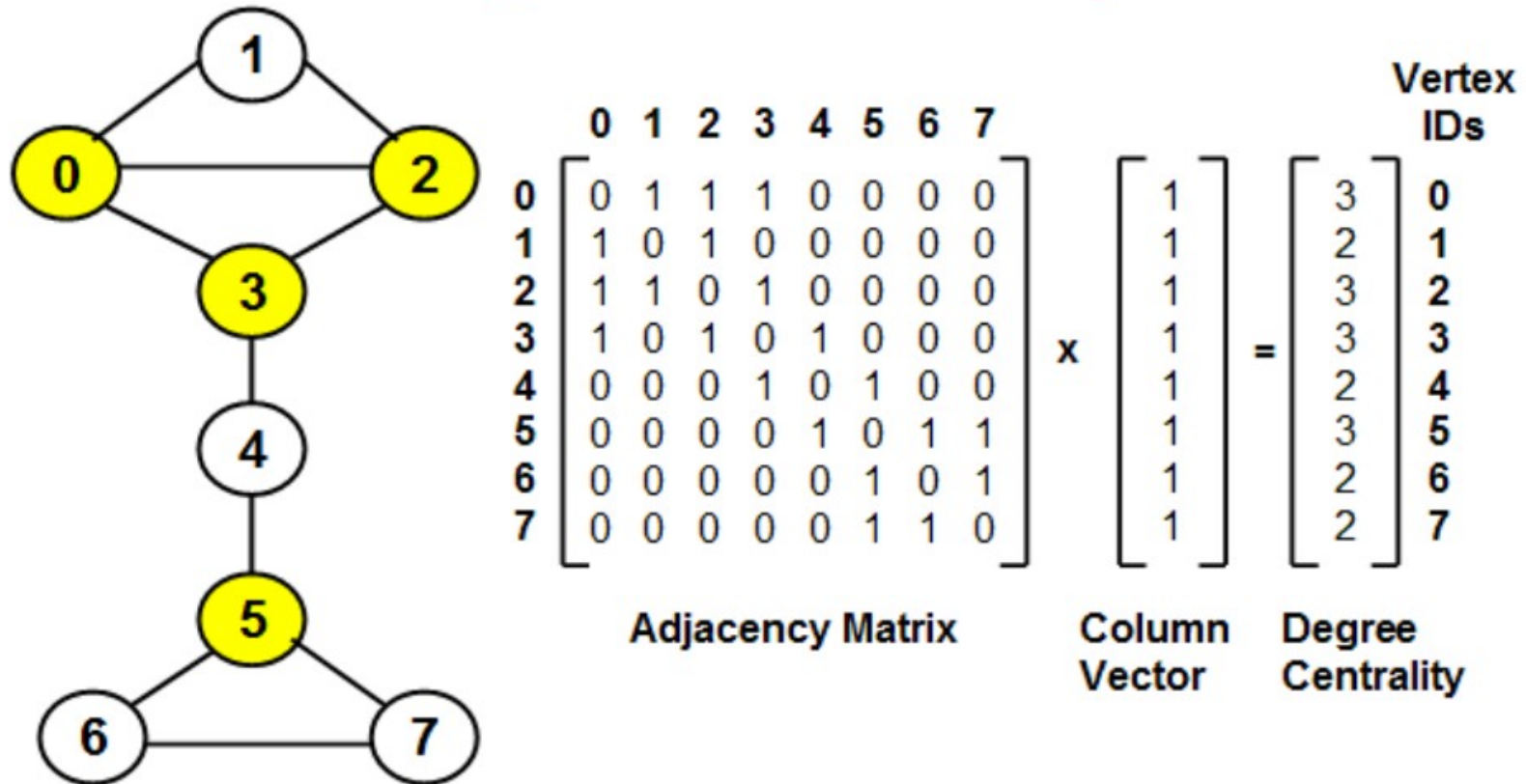
$$C_D(i) := \sum_{j|(i,j) \in E} A_{ij}.$$

- ▶ **High degree centrality** value of a given node
 - ⇒ The node has a **large number of neighbors**
 - ⇒ **Closely related** to its neighbors (in weighted similarity graphs)
- ▶ For directed networks, both in-degree and out-degree centralities
- ▶ Does not capture **cascade effects**
 - ⇒ I am more important if my neighbors are important

(imagens/slides A. Ajorlou, MIT)

Centralidade de grau

Degree Centrality



Weakness: Very likely that more than one vertex has the same degree and not possible to uniquely rank the vertices

(imagens/slides N. Meghanathan)

Centralidade de proximidade

- ▶ How fast **information can spread** from one node to every other node
- ▶ Inverse of the **sum of the shortest path lengths s_G** to other nodes

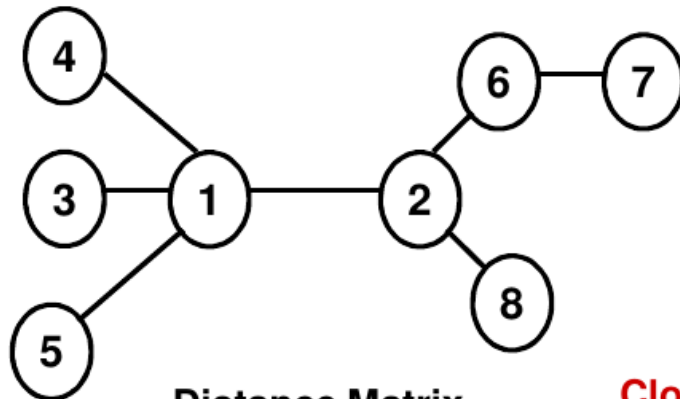
$$C_C(i) := \left(\sum_{j \in V} s_G(i, j) \right)^{-1}.$$

- ▶ Nodes which are close to others on average have high centrality
 - ⇒ Such nodes may have more direct influence on others
- ▶ **Example:** Network of movie actors: two actors are connected if they work together
 - ⇒ Highest centrality 0.4143 for Christopher Lee; Entered into the Guinness Book of World Records in 2007 for most screen credits
 - ⇒ Lowest centrality 0.1154 for Leia Zanganeh, an Iranian film and theater actress
- ▶ Two limitations associated with closeness centrality
 - ⇒ Spans a very small dynamic range
 - ⇒ In disconnected graphs. centrality is zero for all nodes

(imagens/slides A. Ajorlou, MIT)

Centralidade de proximidade

Closeness and Farness Centrality



Distance Matrix

	1	2	3	4	5	6	7	8	
1	0	1	1	1	1	2	3	2	11
2	1	0	2	2	2	1	2	1	11
3	1	2	0	2	2	3	4	3	17
4	1	2	2	0	2	3	4	3	17
5	1	2	2	2	0	3	4	3	17
6	2	1	3	3	3	0	1	2	15
7	3	2	4	4	4	1	0	3	21
8	2	1	3	3	3	2	3	0	17

Closeness
Sum of
distances

Principal
Eigenvalue
 $\eta_1 = 16.315$

Ranking of Nodes

Score	Node ID
0.2518	2
0.2527	1
0.3278	6
0.3763	8
0.3771	3
0.3771	4
0.3771	5
0.4439	7

Farness
Principal
Eigenvector
 $\delta_1 =$
[0.2527

0.2518
0.3771
0.3771
0.3771
0.3771
0.3278
0.4439
0.3763]

(imagens/slides N. Meghanathan)

Centralidade de intermediação

- ▶ Control on the **optimal flow** within a graph
 - ⇒ Nodes that fall within the **shortest path** of many pairs of nodes

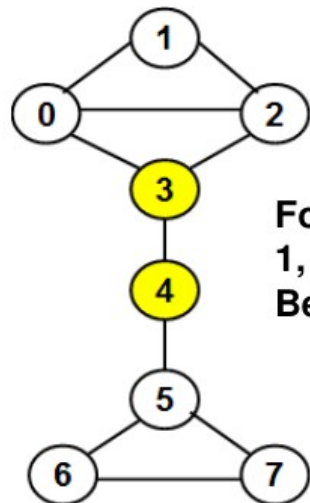
- ▶ The **number of shortest paths** from j to k is σ_{jk}
- ▶ The **number of shortest paths** from j to k **going through** i is $\sigma_{jk}(i)$

$$C_B(i) := \sum_{\substack{j,k \in V \\ j \neq i \neq k}} \frac{\sigma_{jk}(i)}{\sigma_{jk}}.$$

- ▶ Look at the **shortest paths** for every two nodes distinct from i
 - ⇒ Sum the **proportion that contain node i**
- ▶ Betweenness is not a measure of connectivity.
 - ⇒ For the network of movie actors, highest centrality node is Spanish actor Fernando Rey, with centrality of 7.47×10^8
 - ⇒ He starred in many French, American and Spanish movies.

(imagens/slides A. Ajorlou, MIT)

Centralidade de intermediação



For vertices
1, 6 and 7
Betw.C = 0

Betweenness for Vertex 5
 Pair (6,0) --->1 / 1
 Pair (7,0) --->1 / 1
 Pair (6,1) --->2 / 2
 Pair (7,1) --->2 / 2
 Pair (6,2) --->1 / 1
 Pair (7,2) --->1 / 1
 Pair (6,3) --->1 / 1
 Pair (7,3) --->1 / 1
 Pair (6,4) --->1 / 1
 Pair (7,4) --->1 / 1
Total of all Betweenness (Vertex 5): 10

Betweenness for Vertex 0
 Pair (3,1) --->1 / 2
 Pair (4,1) --->1 / 2
 Pair (5,1) --->1 / 2
 Pair (6,1) --->1 / 2
 Pair (7,1) --->1 / 2
Total of all Betweenness (Vertex 0): 2.5

Betweenness for Vertex 2
 Pair (3,1) --->1 / 2
 Pair (4,1) --->1 / 2
 Pair (5,1) --->1 / 2
 Pair (6,1) --->1 / 2
 Pair (7,1) --->1 / 2
Total of all Betweenness (Vertex 2): 2.5

Betweenness for Vertex 3
 Pair (4,0) --->1 / 1
 Pair (5,0) --->1 / 1
 Pair (6,0) --->1 / 1
 Pair (7,0) --->1 / 1
 Pair (4,1) --->2 / 2
 Pair (5,1) --->2 / 2
 Pair (6,1) --->2 / 2
 Pair (7,1) --->2 / 2
 Pair (4,2) --->1 / 1
 Pair (5,2) --->1 / 1
 Pair (6,2) --->1 / 1
 Pair (7,2) --->1 / 1
Total of all Betweenness (Vertex 3): 12

Betweenness for Vertex 4
 Pair (5,0) --->1 / 1
 Pair (6,0) --->1 / 1
 Pair (7,0) --->1 / 1
 Pair (5,1) --->2 / 2
 Pair (6,1) --->2 / 2
 Pair (7,1) --->2 / 2
 Pair (5,2) --->1 / 1
 Pair (6,2) --->1 / 1
 Pair (7,2) --->1 / 1
 Pair (5,3) --->1 / 1
 Pair (6,3) --->1 / 1
 Pair (7,3) --->1 / 1
Total of all Betweenness (Vertex 4): 12

(imagens/slides N. Meghanathan)

Centralidade de autovetor

- ▶ **Linear combination** of the centrality of the neighbors
⇒ **Few important** neighbors can weigh more than **many unimportant** ones

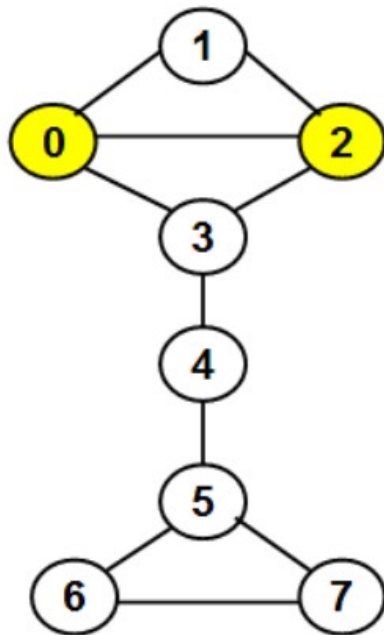
$$C_E(i) := \frac{1}{\lambda} \sum_{(i,j) \in E} A_{ij} C_E(j),$$

- ▶ This definition leads to an **eigenvector** problem
- ▶ C_E is the **dominant eigenvector** of the adjacency matrix
⇒ Perron-Frobenius guarantees **all positive elements**

(imagens/slides A. Ajorlou, MIT)

Centralidade de autovetor

Eigenvector Centrality (2)



Iteration 3

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.428 \\ 0.321 \\ 0.428 \\ 0.428 \\ 0.321 \\ 0.321 \\ 0.268 \\ 0.268 \end{bmatrix} = \begin{bmatrix} 1.177 \\ 0.856 \\ 1.284 \\ 1.177 \\ 0.749 \\ 0.857 \\ 0.589 \\ 0.589 \end{bmatrix} \equiv \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix}$$

Normalized Value = 2.67

Iteration 4

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix} = \begin{bmatrix} 1.243 \\ 0.922 \\ 1.203 \\ 1.203 \\ 0.762 \\ 0.723 \\ 0.542 \\ 0.542 \end{bmatrix} \equiv \begin{bmatrix} 0.471 \\ 0.349 \\ 0.456 \\ 0.456 \\ 0.289 \\ 0.274 \\ 0.205 \\ 0.205 \end{bmatrix}$$

Normalized Value = 2.64

After 7 iterations

Vertex ID	0	1	2	3	4	5	6	7
Principal Eigenvector	0.489	0.364	0.489	0.467	0.264	0.232	0.155	0.155

(imagens/slides N. Meghanathan)

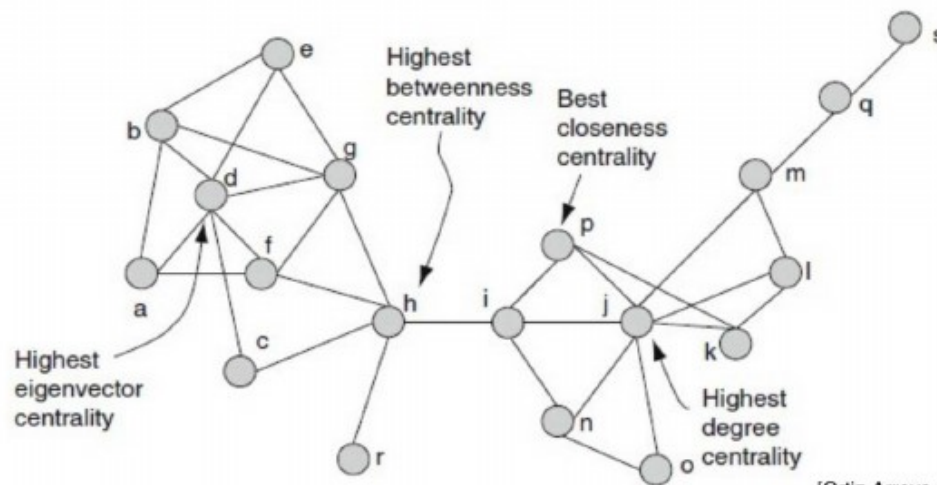
Interpretações e comparações

- ▶ What would **degree**, **closeness**, and **betweenness** centrality reveal?
- ▶ **Degree** $\Rightarrow$ Most friends $\Rightarrow$ Most **popular** person
- ▶ **Closeness** $\Rightarrow$ Can quickly reach the whole group (directly or indirectly)
 $\Rightarrow$ Relevant if we want to **quickly spread information** in the network
- ▶ **Betweenness** $\Rightarrow$ Power in the transmission of information
 $\Rightarrow$ Relevant if we want to **influence communication** between groups
- ▶ All of them are right, they just reveal different features

(imagens/slides A. Ajorlou, MIT)

Interpretações e comparações

- ▶ Different measures target different notions of importance
- ▶ In a friendship network, degree centrality would correspond to who is the most popular kid.
- ▶ Closeness centrality would correspond to who is closest to the rest of the group,. Useful for spreading information
- ▶ Betweenness would tell us about graph “cut points”, edges whose deletion will cause multiple connected components



[Ortiz-Arroyo 10]

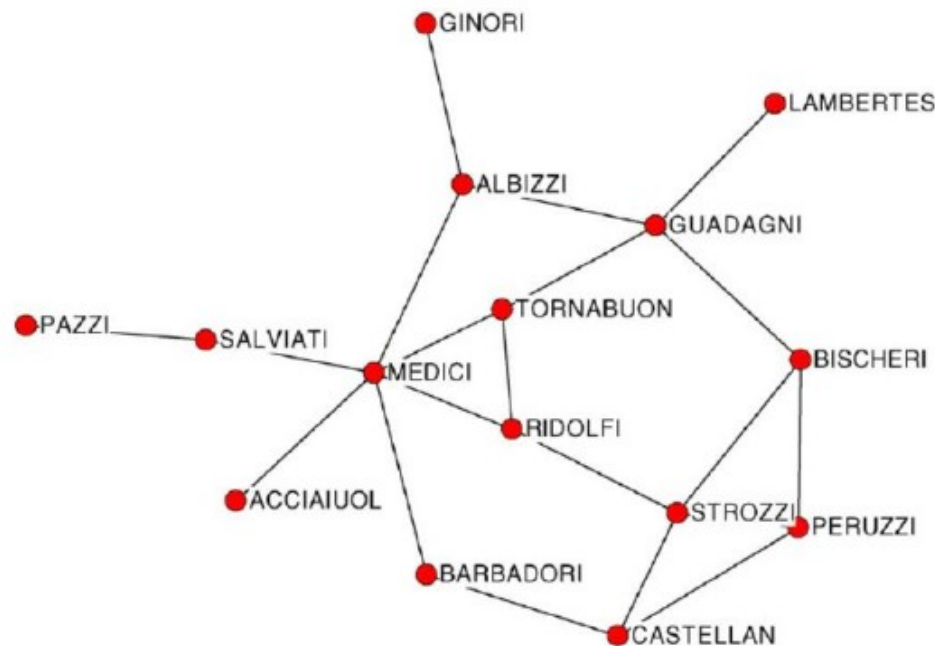
Ortiz-Arroyo, Daniel. "Discovering Sets of Key Players in Social Networks." Chapter 2 in *Computational Social Network Analysis: Trends, Tools and Research Advances*. Springer, 2009. © Springer. All rights reserved. This content is excluded from our Creative Commons license.

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Interpretações e comparações

- Why were the Medicis the most influential family in 15th c. Florence?
- Political and friendship structure [Padgett & Ansell 93]



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Source: Stephen P. Borgatti. "Centrality and Network Flow." *Social Networks* 27 (2005): 55–71.

- Highest **betweenness** centrality by far
⇒ Part of many deals between families supported by marriage linkages

(imagens/slides A. Ajorlou, MIT)

Referências Bibliográficas

- Bron, Coen; Kerbosch, Joep (1973), "Algorithm 457: finding all cliques of an undirected graph", *Commun. ACM*, ACM, 16 (9):575–577, doi:10.1145/362342.362367.
- Sedgewick, R. & Wayne, K. *Algorithms* (4th ed.), Addison-Wesley, 2011.
- Wilson, R. & Watkins, J. *Graphs: An Introductory Approach*. John Wiley, 1990.
- Ziviani, N. *Projeto de algoritmos com implementações em pascal e C*. Cengage, 2011.